RailFlow

RailFlow: End-to-End Product Analytics

Optimizing Train Booking Conversion with Machine Learning & Real-Time Incentives.

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The Challenge: High Traffic, Low Conversion.

Bullet Points:

- **Context:** Users frequently search for routes (e.g., Paris → Lyon) but abandon the funnel before payment.
- **The Gap:** No visibility into *why* users drop off and no mechanism to re-engage them in real-time.
- **The Goal:** Build a system to identify "High Intent" users and trigger targeted incentives.

Modern Data Stack Principles

Ingestion	Python (`Faker` , `Polars`)	Generating 10k+ realistic synthetic log rows with schema enforcement.
Privacy	Python (`hashlib`)	**GDPR Compliance:** SHA-256 hashing of PII (emails) before warehousing.
Warehousing	Google BigQuery	Serverless Data Warehouse located in `europe-west9` (Paris).
Transformation	dbt (Data Build Tool)	Modeling raw data into Star Schema (`fct_daily_performance` , `dim_users`).
Machine Learning	Scikit-Learn	Random Forest Classifier to predict booking probability.
Application	Streamlit	Interactive web app for stakeholders to test the ML model.
Reverse ETL	Python	Syncing "High Churn Risk" segments to Marketing tools (simulated).
Validation	SciPy	A/B Testing statistical significance calculator.

Building a GDPR-Compliant Foundation.

01

Ingestion: Python pipeline generating 10k+ realistic synthetic log events (Searches, Bookings).

02

Privacy First: Implemented **SHA-256 Salting & Hashing** for user emails before data enters the warehouse.

```python
# 1. The Hashing Function
# I used SHA-256, the Industry standard

def has_pii(email):
  # Adding a 'salt' (a secret random string) to prevent rainbow table attacks
  salt = 'SML7_SECRET_S6LT_28241'
  salted_email = email + salt
  return hashlib.sha256(salted_email.encode()).hexdigest()

  # Simulate Raw Data
raw_users = pd.DataFreme({
    'user_id': [1,3,3],
    'name': ['Jeon Dupont', 'Marie Curie', 'Pierre Avoine'],
    'email': ['jean.dupont@gmail.com', 'marie.curie@yahoo.fr', 'pierre@sncf.fr']
})

print("📮 RAW DATA (Internal Only):")
pr1nt(raw_users)
```

```
📮 RAN DATA (Internal Only):
   user_id           name                 email
0        1    Jean Dupont  jean.dupont@gmail.com
1        2    Marie Curie   marie.corle@yehoe.fr
2        3  Pierre Avoine         pierre@sncf.fr
```

```python
# 3. Apply Anonymization
raw_users['email_hash'] = raw_users['email'].apply(has_pii)
```

```python
# Drop the PII columns
clean_users = raw_users.drop(columns=['email','name'])
```

```python
print("\n🔒 ANONYMIZED DATA (Ready for Warehouse):")
display(clean_users)
```

🔒 ANONYMIZED DATA (Ready for Warehouse):

	user_id	email_hash
0	1	f4b77b2ef6a8ae4c63b1a6474b2edc2ddd7B0da2d24ce4...
1	2	4432c2d4f2ef90e741511139a100ec98792d41f3f3f4cd...
2	3	1d94daa78b5c7a677c9455bf300be8728b8aeaa8074491...

Modeling the User Journey

Strategy: Dimensional Modeling (Star Schema) in BigQuery.

Key Model: `int_search_bookings` – Joins search logs with transaction data to calculate conversion rates per route.

Outcome: A trusted "Single Source of Truth" for the BI team.

Predicting User Intent

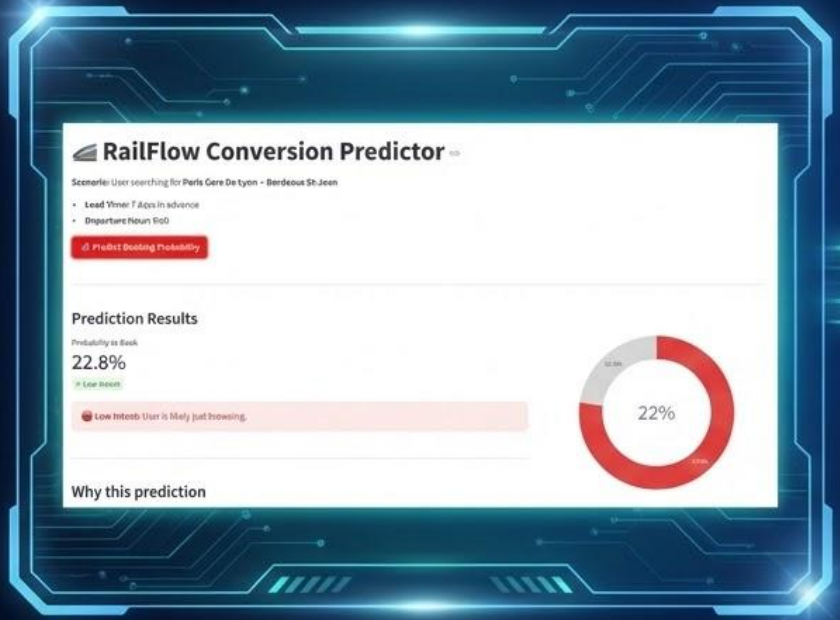
Model: Random Forest Classifier (Scikit-Learn).

Objective: Predict is_converted (0 or 1).

Key Insight: Lead Time (days before travel) was the #1 predictor of conversion.

The "Smart Discount" Engine

LIVE APPLICATION (Streamlit)



ACTION

Deployed a live Streamlit application for Product Managers.

LOGIC FLOW

High Probability:
Do Nothing (Save Margin).

Medium Probability (30-70%):
Trigger 5% Discount.

Low Probability:
Ignore.

Prioritization & Requirements

RICE Framework: Selected this feature (Score: 6,000) over 'Social Login' due to higher revenue impact.

PRD: Defined 'Medium Intent' thresholds to prevent revenue cannibalization.

Feature Prioritization Data

⚖️ Feature Prioritization (RICE Framework)

We evaluated 3 potential features for Q4. The **Smart Discount Engine** was selected as the winner.

The Formula

$$RICEScore = \frac{(Reach \times Impact \times Confidence)}{Effort}$$

The Evaluation

Feature Idea	R(sach) (Users/Month)	I(mpact) (1-3 Scale)	C(onfidence) (%)	E(ffort) (Person-Weeks)	FINAL SCORE	Decision
Smart Discount Engine	100,000	3 (High)	80%	4 wks	**6,000**	✅ **BUILD**
Social Login (Google/FB)	100,000	1 (Low)	90%	2 wks	**4,500**	1️⃣ Backlog
Carbon Footprint Calculator	100,000	0.5 (Min)	100%	3 wks	**1,666**	❌ Kill

Rationale

1. **Smart Discount Engine:** Although Effort is higher (4 weeks), the Revenue Impact (3) is massive compared to a simple UX tweak like Social Login.
2. **Social Login:** Good for onboarding, but doesn't drive immediate revenue.
3. **Carbon Calculator:** High user request, but low business impact. Deferred to Q1.

Proving the Value

```
...
✅ RESULT: STATISTICALLY SIGNIFICANT!
Conclusion: The 0.5% lift is real. We should launch the feature.
```

Experimentation:
Designed an A/B Test (Control vs. Variant).

Power Analysis:
Calculated required sample size (~86k users) to prove a 0.5% lift.

Activation:
Reverse ETL pipeline to sync “High Churn Risk” users to Braze/Salesforce

Summary & Next Step

Result:
A full-cycle data product: From raw logs to revenue-generating actions.

Next Steps:
Deploy model to Airflow for daily batch predictions.

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GitHub Repository: RailFlow